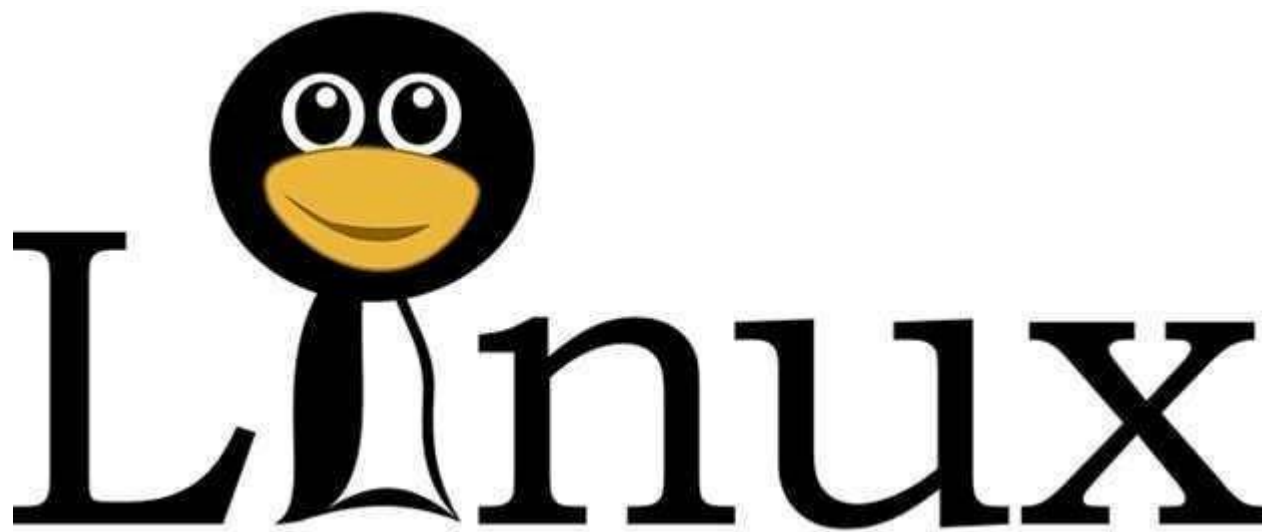




Assignment - 2

Points Covered in this Assignment-2

1. User Administration

- Authentication, Authorization, and Auditing

2. Commands to Learn

- useradd, passwd, userdel, usermod
- groupadd, groupdel, groupmod
- su and su -with examples

3. User and Group Information

- User and group information files
- Password information files

4. Password Policies

- chage command and its options

5. User Monitoring and Auditing

- Commands: w, last, lastb

6. Sudo Power

- wheel group

7. Default Configuration Files

- /etc/default/useradd
 - /etc/login.defs
 - /etc/security/limits.conf
-

1. Create some users:

- Named **"alex"** with its home directory at **/home/user1** and give password **"pass1"**.
- Named **"brew"** with its home directory at **/mnt/user2** and give password **"pass2"**.
- Named **"nora"** without its home directory
- Named **"panny"** with custom UID **2112**, and assign password **"pass-4"**
- Named **'texas'** without using the **useradd** or **adduser** commands.

**(Hint: Make changes in the 7 user configuration files)*

```
[root@localhost ~]# sudo useradd -m -d /home/user1 alex
echo "pass1" | sudo passwd --stdin alex
sudo useradd -m -d /home/user1 alex
echo "pass1" | sudo passwd --stdin alex
Changing password for user alex.
passwd: all authentication tokens updated successfully.
useradd: user 'alex' already exists
Changing password for user alex.
passwd: all authentication tokens updated successfully.
[root@localhost ~]# sudo useradd -m -d /mnt/user2 brew
echo "pass2" | sudo passwd --stdin brew
Changing password for user brew.
passwd: all authentication tokens updated successfully.
[root@localhost ~]# sudo useradd -M nora
[root@localhost ~]# sudo useradd -u 2112 panny
echo "pass-4" | sudo passwd --stdin panny
Changing password for user panny.
passwd: all authentication tokens updated successfully.
[root@localhost ~]# sudo vim /etc/passwd
[root@localhost ~]# openssl passwd -6
```

2. Log in as user alex using the **su** and **su -** commands, and explain their differences

su command switches to the user alex while keeping the current environment variables.

su - command switches to alex and loads their complete login environment, including paths and configurations.

Commands:

su alex

su - alex

3. Set a password policy for all above users with the following requirements:

- The maximum password age should be 30 days, and the minimum password age should be 10 days.
- Set the password expiry date for all users to December 31, 2025.

```
[root@localhost ~]# sudo chage --maxdays 30 --mindays 10 john
[root@localhost ~]# sudo chage --expiredate 2025-12-31 john
[root@localhost ~]# sudo chage -l john
Last password change           : Jan 30, 2025
Password expires                : Mar 01, 2025
Password inactive               : never
Account expires                 : Dec 31, 2025
Minimum number of days between password change : 10
Maximum number of days between password change : 30
Number of days of warning before password expires : 7
[root@localhost ~]# █
```

4. Modify the user "alex":

- Add a comment: "I am alex"
- Change the UID to 2581
- Change the shell to "nologin"

```
[root@localhost ~]# id alex
uid=2581(alex) gid=4099(alex) groups=4099(alex)
[root@localhost ~]# sudo chown -R alex:alex /home/alex
[root@localhost ~]# ls -ld /home/alex
drwx-----. 3 alex alex 78 Jan 30 22:55 /home/alex
[root@localhost ~]# sudo usermod -s /sbin/nologin alex
usermod: no changes
[root@localhost ~]# grep alex /etc/passwd
alex:x:2581:4099:I am alex:/home/alex:/sbin/nologin
[root@localhost ~]# id alex
grep alex /etc/passwd
ls -ld /home/alex
uid=2581(alex) gid=4099(alex) groups=4099(alex)
alex:x:2581:4099:I am alex:/home/alex:/sbin/nologin
drwx-----. 3 alex alex 78 Jan 30 22:55 /home/alex
[root@localhost ~]#
```

5. Create group with following configuration:

- Named "north" with secondary group member "alex" & "texas".
- Named "south" with GID "2222".

```
[root@localhost ~]# sudo groupadd north
[root@localhost ~]# sudo useradd alex
sudo useradd texas
[root@localhost ~]# sudo groupadd -g 2222 south
[root@localhost ~]# grep north /etc/group
grep south /etc/group
id alex
id texas
north:x:4100:alex
south:x:2222:
uid=2581(alex) gid=4099(alex) groups=4099(alex),4100(north)
```

6. Grant user **Alex** administrative privileges through the wheel group so that Alex can add Panny to the admin group without requiring root access.

```
[root@localhost ~]# sudo useradd Alex
[root@localhost ~]# sudo usermod -aG wheel Alex
[root@localhost ~]# sudo visudo
[root@localhost ~]# su - Alex
[Alex@localhost ~]$ sudo usermod -aG wheel Panny
usermod: user 'Panny' does not exist
[Alex@localhost ~]$ sudo useradd Panny
[Alex@localhost ~]$ sudo passwd Panny
Changing password for user Panny.
New password:
BAD PASSWORD: The password is a palindrome
Retype new password:
passwd: all authentication tokens updated successfully.
```

7. Change the group name from "south" to "dakshin".

```
[root@localhost ~]# sudo groupmod -n dakshin south
[root@localhost ~]# getent group dakshin
dakshin:x:2222:
[root@localhost ~]#
```

8. Create a system user named **"ping"** and check its UID.

```
[root@localhost ~]# sudo useradd ping
[root@localhost ~]# id ping
uid=4101(ping) gid=4103(ping) groups=4103(ping)
[root@localhost ~]#
```

9. Create a group named **goa** with GID 11000. Set this group as the supplementary group for **"brew"**

```
[root@localhost ~]# sudo groupadd -g 11000 goa
[root@localhost ~]# sudo usermod -aG goa brew
usermod: user 'brew' does not exist
[root@localhost ~]# sudo useradd brew
Creating mailbox file: File exists
[root@localhost ~]# sudo usermod -aG goa brew
[root@localhost ~]#
```

10. Create a group named **"prod"**. Then, create two users, user2 and user1, and set both the user's primary group to **prod**.

```
[root@localhost ~]# sudo groupadd prod
[root@localhost ~]# sudo useradd -g prod user1
useradd: user 'user1' already exists
[root@localhost ~]# sudo useradd -g prod user2
[root@localhost ~]# id user1
id user2
uid=4097(user1) gid=4097(user1) groups=4097(user1)
uid=4103(user2) gid=11002(prod) groups=11002(prod)
[root@localhost ~]#
```

11. Change the password policy for the USER3 and USER4 accounts to expire on 2026-01-15.

```
[root@localhost ~]# sudo useradd USER3
[root@localhost ~]# sudo chage -E 2026-01-15 USER3
[root@localhost ~]# sudo useradd USER4
[root@localhost ~]# sudo chage -E 2026-01-15 USER4
[root@localhost ~]# sudo chage -l USER3
sudo chage -l USER4
Last password change          : Jan 30, 2025
Password expires              : never
Password inactive             : never
Account expires               : Jan 15, 2026
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
Last password change          : Jan 30, 2025
Password expires              : never
Password inactive             : never
Account expires               : Jan 15, 2026
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
[root@localhost ~]#
```

12. **Configure administrative rights** for all members of the **Goa** group to execute any command as any user.

```
[root@localhost ~]# sudo visudo
[root@localhost ~]# sudo whoami
root
[root@localhost ~]#
```

13. How would you check all failed login attempts on the system from the last 10 days? Write the command and display the output

To view failed login attempts from the last 10 days:

```
[root@localhost ~]# sudo journalctl _COMM=sshd -p auth.err --since "10 days ago"
Unknown log level auth.err
[root@localhost ~]#
```

14. How would you determine how many users are currently logged into the system? Write the command to achieve this.

```
[root@localhost ~]# who
root      seat0      2025-01-31 00:07 (login screen)
root      tty2       2025-01-31 00:07 (tty2)
[root@localhost ~]# who | wc -l
2
[root@localhost ~]#
```

15. Add the user **"sara"** to the **"wheel"** group and create a collaborative directory **/collaborative/infodir**

```
[root@localhost ~]# sudo useradd sara
[root@localhost ~]# sudo usermod -aG wheel sara
[root@localhost ~]# sudo mkdir -p /collaborative/infodir
[root@localhost ~]# sudo chgrp wheel /collaborative/infodir
sudo chmod 770 /collaborative/infodir
[root@localhost ~]#
```

16. **Configure login/logout messages:**

- When you log in with a new user, display a message: "Hello, you are logged in as USER" (where USER is replaced with the logged-in username).
- When you log out, display: "You are logged out now".

Using commands:

```
sudo nano /etc/profile
```

```
sudo nano /etc/bash.bash_logout
```

17. Configure system parameters for newly created users:Warning period for password expiry: 5 days

- Minimum user UID: 2000
- Maximum user UID: 70000

```
[root@localhost ~]# sudo vi /etc/login.defs
[root@localhost ~]#
```

18. **Create a directory /data** and configure the system so that all newly created users get /data as their home directory by default.

```
[root@localhost ~]# sudo mkdir /data
[root@localhost ~]# sudo chmod 777 /data
[root@localhost ~]# sudo vi /etc/default/useradd
[root@localhost ~]#
```

19. Name a file where we can set a file size limit upto 200 MB for a single file.

```
[root@localhost ~]# sudo nano /etc/security/limits.conf
[root@localhost ~]#
```

20. **Check the last three users** who logged into your system

```
[root@localhost ~]# last -n 3
root      tty2          tty2          Fri Jan 31 00:07    still logged in
root      seat0         login screen   Fri Jan 31 00:07    still logged in
reboot    system boot   5.14.0-362.8.1.e Fri Jan 31 00:06    still running

wtmp begins Thu Jan 23 12:05:30 2025
[root@localhost ~]#
```

21. As a system administrator, how would you configure the system to ensure that:

- Automatically create an instructions.txt file in the home directory of every new user upon account creation.
- Ensure that the mail directory for every newly created user is set to /home/spool/mail/ by default?"

```
[root@localhost ~]# sudo echo "Welcome to your new account!" > /etc/skel/instructions.txt
[root@localhost ~]# ls /etc/skel
instructions.txt
[root@localhost ~]# sudo vi /etc/default/useradd
[root@localhost ~]#
```

22. **Delete some users**

- Named **'alex'** and **'brew'** with its all data contents including mail data.

```
[root@localhost ~]# ls -l /var/mail/
total 0
-rw-rw----. 1 4094 mail 0 Jan 26 23:30 alex
-rw-rw----. 1 Alex mail 0 Jan 30 23:36 Alex
-rw-rw----. 1 4095 mail 0 Jan 26 23:31 brew
-rw-rw----. 1 deep mail 0 Jan 24 09:37 deep
-rw-rw----. 1 4092 mail 0 Jan 25 10:30 harry
-rw-rw----. 1 john mail 0 Jan 30 22:45 john
-rw-rw----. 1 Molly mail 0 Jan 23 12:06 Molly
-rw-rw----. 1 4090 mail 0 Jan 24 09:36 neel
-rw-rw----. 1 nora mail 0 Jan 26 23:31 nora
-rw-rw----. 1 2112 mail 0 Jan 26 23:32 panny
-rw-rw----. 1 Panny mail 0 Jan 30 23:41 Panny
-rw-rw----. 1 ping mail 0 Jan 31 00:10 ping
-rw-rw----. 1 4093 mail 0 Jan 25 10:30 raj
-rw-rw----. 1 sara mail 0 Jan 31 00:37 sara
-rw-rw----. 1 user1 mail 0 Jan 27 09:52 user1
-rw-rw----. 1 user2 mail 0 Jan 31 00:16 user2
-rw-rw----. 1 USER3 mail 0 Jan 31 00:19 USER3
-rw-rw----. 1 USER4 mail 0 Jan 31 00:20 USER4
[root@localhost ~]# sudo rm /var/mail/alex
sudo rm /var/mail/brew
[root@localhost ~]# sudo rm -rf /home/alex
sudo rm -rf /home/brew
[root@localhost ~]# sudo userdel alex
sudo userdel brew
userdel: user 'alex' does not exist
userdel: user 'brew' does not exist
[root@localhost ~]#
```